

ALEX ARIZA HERRERA

AI Engineer | Production AI Systems | Full-Stack Development

Based in Colombia | Remote-First & Onsite Opportunities | arizah2020@gmail.com

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Professional Summary

AI Engineer focused on building production-ready LLM and RAG systems across regulated domains and knowledge-intensive applications. Experienced in developing end-to-end AI products, including retrieval pipelines, document processing, vector search, observability, and full-stack web platforms. Built healthcare software with compliance, auditing, and interoperability requirements, as well as enterprise and local-first knowledge assistants powered by modern LLM architectures.

Selected Projects

Clinic Operations Copilot

[View Project](#)

Regulated healthcare platform for dental clinics in Colombia, designed to support electronic health records, compliance, auditing, and future AI-assisted workflows.

- Designed and developed a regulatory-compliant Electronic Health Record (EHR) platform for dental clinics, addressing healthcare data governance, privacy, auditability, and interoperability requirements under Colombian healthcare regulations.
- Implemented immutable, append-only clinical record workflows where signed entries cannot be modified or deleted; amendments generate linked correction records that preserve chronological integrity, legal traceability, and historical accuracy.
- Engineered compliance-critical capabilities including clinical record signing and closure, versioned consent management, comprehensive audit trails, fine-grained RBAC, and privacy-by-design safeguards for sensitive health information.
- Developed healthcare interoperability workflows aligned with HL7 FHIR and Colombia's RDA (Digital Healthcare Summary) framework, enabling standardized data exchange with Ministry of Health platforms.
- Architected asynchronous processing and event-driven workflows using transactional outbox patterns and background workers, establishing integration points for future AI capabilities while enforcing strict isolation of legally protected clinical records.

Stack: Python 3.12, FastAPI, SQLAlchemy Async, Alembic, PostgreSQL, Redis, Celery, Next.js, TypeScript, Docker, OpenAPI, GitHub Actions, AWS EC2/S3/IAM/CloudWatch, HL7 FHIR, RDA

AI System for SGR/MGA Project Formulation — Multi-Agent Planning Assistant

[View Project](#)

Designed an AI-assisted workflow system for transforming public-sector requirements into structured pre-investment planning artifacts aligned with Colombia's SGR/MGA methodology.

- Designed multi-step AI workflows to decompose complex planning processes into problem definition, population analysis, territorial assessment, alternative evaluation, gap identification, and roadmap generation.
- Developed agent orchestration workflows in which a coordinating agent delegated specialized planning tasks to domain-specific AI agents, enabling traceable generation of structured planning artifacts.
- Implemented evidence-first reasoning pipelines where recommendations, assumptions, requirements, and proposed interventions remain linked to supporting documents, references, or explicitly identified information gaps.
- Built structured knowledge-generation systems that transform unstructured stakeholder requirements into auditable outputs suitable for public-sector planning and decision-support workflows.
- Designed human-in-the-loop validation mechanisms to preserve accountability, transparency, and expert oversight while preventing autonomous approval or prioritization decisions.

Stack: LLM Workflows, Agent Orchestration, Structured Prompting, Knowledge Modeling, Document Analysis, Human-in-the-Loop Systems, Domain-Driven AI Workflows

AI Internal Knowledge Copilot — Enterprise RAG Application

[View Project](#)

Built an AI-powered knowledge assistant that delivers verifiable, citation-backed answers from internal documentation.

- Developed an end-to-end RAG application with Next.js, NestJS, PostgreSQL, authentication, document ingestion, and conversational querying capabilities.
- Implemented document processing pipelines with chunking, normalization, metadata extraction, embeddings, and vector search to ground LLM outputs in uploaded content.
- Improved retrieval quality through adaptive chunking strategies and structured metadata, reducing hallucinations and increasing answer relevance.
- Managed the complete product lifecycle including architecture, feature planning, Git workflows, deployment environments, testing, and iterative releases.

Stack: TypeScript, Next.js, React, NestJS, PostgreSQL, Prisma, Docker, AWS, LangChain, OpenAI API

Local RAG Knowledge Copilot — Local-First AI Retrieval Platform

[View Project](#)

Designed a privacy-focused RAG platform for private document collections using fully local retrieval and inference pipelines.

- Built a local-first document intelligence pipeline with PDF ingestion, deterministic hashing, duplicate detection, and traceable metadata at document, page, and chunk levels.
- Implemented hybrid retrieval using ChromaDB, SQLite FTS5, and Reciprocal Rank Fusion (RRF) to combine semantic similarity with exact lexical matching.
- Integrated local embedding generation and inference using sentence-transformers and llama.cpp, eliminating dependency on external AI providers.
- Developed citation-backed answer generation workflows with retrieval diagnostics and transparent source attribution.
- Engineered observability capabilities including structured logging, correlation IDs, Prometheus metrics, and latency monitoring across ingestion, retrieval, and generation stages.

Stack: Python 3.12, FastAPI, Pydantic v2, Docling, ChromaDB, SQLite FTS5, sentence-transformers, llama.cpp, structlog, prometheus-client, React 19, TypeScript, Vite, Tailwind CSS, Radix UI, TanStack Query

Education

Universidad de los Andes, Electronic Engineering Degree

The University of Texas at Austin, Certificate in Full Stack Software Development: Building Scalable Cloud Applications

Skills

Languages: Python, TypeScript, SQL

AI & LLM Systems: RAG, Vector Search, Embeddings, Hybrid Retrieval, Prompt Engineering, LangChain, Local LLM Inference (llama.cpp), Agentic Workflows

Backend: FastAPI, NestJS, PostgreSQL, SQLAlchemy, Prisma, Redis, REST APIs, Authentication, RBAC

Frontend: React, Next.js, TypeScript, Tailwind CSS, TanStack Query

Infrastructure: Docker, AWS, GitHub Actions, CI/CD, Observability, Prometheus